

Parameter-disturbance robustness of the exact DCT-IV init

Controlled O(2)-twiddle DCT-IV on DIV2K-8q: Gaussian jitter ($\sigma = 0.1$) of a random fraction of the 2200 exact-init gate entries, 1008-step top-10% training, 3 seeds

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Setup

Starting from the **exact** analytic DCT-IV (`DCT4Basis(8, 8, parametrization: "controlled")` — the O(2)-twiddle form of pdf PR #24, commit 5365a5a, whose CRY twiddle is applied to the control = 1 half only), we perturb a random fraction f of its **2200 real gate-tensor entries** with on-manifold Gaussian jitter: add $N(0, 0.1)$ to the selected entries and re-project each touched gate onto its manifold (nearest-orthogonal O(4)/O(2) via SVD; the Δ -sign gate by a phase jitter), so the perturbed init stays a valid real-orthogonal DCT-IV and its untrained PSNR is interpretable. The disturbed counts are $\lfloor f \cdot 2200 \rfloor = \{2, 4, 11, 22, 44, 110, 220, 440, 1100, 2200\}$ entries for $f \in \{0.1, 0.2, 0.5, 1, 2, 5, 10, 20, 50, 100\}\%$; at $f = 100\%$ **every** gate is jittered.

Each perturbed init (3 seeds per f) is trained for 1008 steps of top-10% MSE on the full 500-image DIV2K-8q pool (mini-batch 50, a **fixed** shuffle seed so only the perturbation varies), then scored for test PSNR at four keep ratios ρ on the fixed canonical 50-image test set. $f = 0$ is the undisturbed exact-init reference (its untrained PSNR reproduces the classical DCT-IV: 30.54 dB @ $\rho = 0.20$).

Disturbance procedure

The jitter treats the exact DCT-IV's 214 gate tensors as one flat vector of 2200 real entries and acts **per gate**. For disturbance rate f and jitter scale σ :

```
select round(f * 2200) of the 2200 real gate entries, uniform, no replacement
for each gate G that owns >= 1 selected entry:
  add N(0, sigma) to G's selected entries only          # Gaussian jitter
  re-project the noised gate back onto its manifold:
    (2,2,2,2) mirror-U4    -> nearest orthogonal, SVD polar U V^T  [O(4)]
    (2,2) Delta-sign gate  -> phi <- pi + sigma*z ; controlled_phase_diag(phi)
    (2,2) rotation / H     -> nearest orthogonal, SVD polar U V^T  [O(2)]
gates with no selected entry are copied unchanged      # f = 0 = identity
```

The noise is plain additive i.i.d. Gaussian $N(0, \sigma)$ on the **selected raw entries only**; the per-gate **re-projection** is what turns it into an on-manifold perturbation. Each gate is real-orthogonal, so the nearest valid gate to the noised one is its SVD polar factor UV^T (drop the singular values) — this keeps the operator a genuine real-orthogonal DCT-IV, so the perturbed init's untrained PSNR is meaningful rather than an artefact of a non-orthogonal matrix. The Δ -sign gate is not a rotation but a phase stub (its lower-right entry is $e^{i\varphi}$, $\varphi = \pi$ at init), so it is jittered in its phase φ instead. Because a gate moves only when one of its entries is selected, small f nudges a few gates while $f = 100\%$ jitters **every** gate; the **magnitude** each touched gate travels is set by σ , not by f .

Final PSNR vs. disturbance

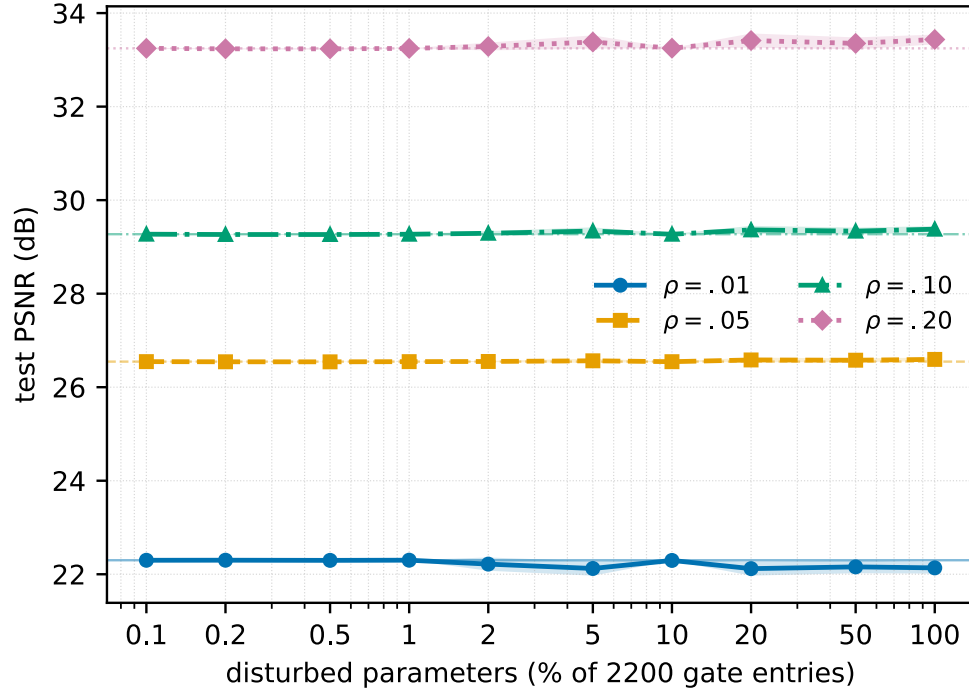


Figure 1: Trained test PSNR vs. the fraction of exact-init parameters disturbed (log-x), one curve per keep ratio ρ , mean $\pm\sigma$ over 3 seeds (shaded). Thin horizontal lines mark the undisturbed exact-init-trained PSNR per ρ .

disturbed	$\rho\{=\}.01$	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.20$
0 (exact)	22.3	26.55	29.27	33.25
0.1%	22.3 ± 0	26.55 ± 0	29.27 ± 0.01	33.25 ± 0.01
0.2%	22.3 ± 0	26.54 ± 0	29.27 ± 0.01	33.24 ± 0.01
0.5%	22.3 ± 0	26.54 ± 0.01	29.26 ± 0.01	33.24 ± 0.02
1%	22.3 ± 0	26.55 ± 0	29.27 ± 0	33.24 ± 0
2%	22.22 ± 0.14	26.55 ± 0.01	29.29 ± 0.05	33.29 ± 0.08
5%	22.12 ± 0.15	26.56 ± 0.04	29.34 ± 0.08	33.38 ± 0.14
10%	22.3 ± 0.01	26.55 ± 0	29.27 ± 0	33.25 ± 0
20%	22.12 ± 0.15	26.58 ± 0.05	29.36 ± 0.09	33.41 ± 0.15
50%	22.16 ± 0.14	26.58 ± 0.04	29.34 ± 0.08	33.35 ± 0.13
100%	22.14 ± 0.12	26.59 ± 0.04	29.38 ± 0.07	33.44 ± 0.11

Trained test PSNR (dB), mean $\pm\sigma$ over 3 perturbation seeds; row 0 (exact) is the undisturbed exact-init-trained reference.

Recovery: perturbed init vs. trained

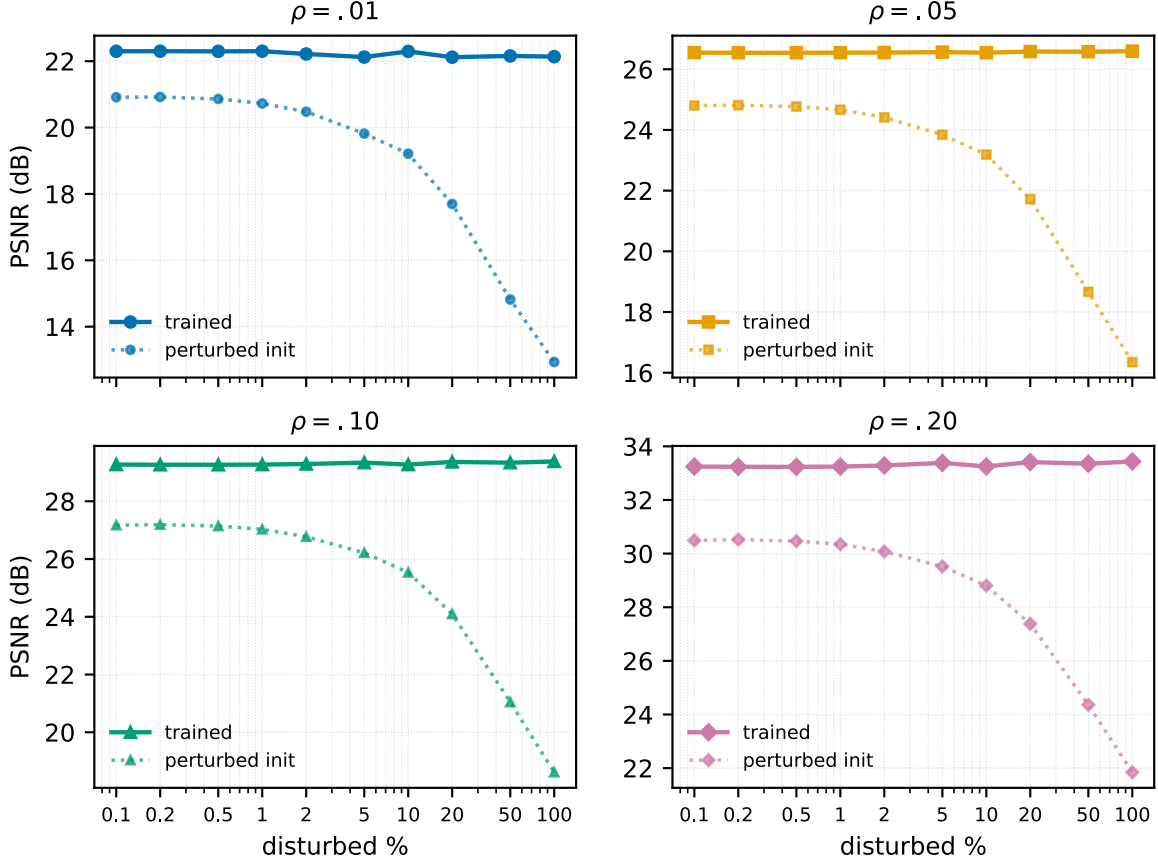


Figure 2: Per- ρ panels: the perturbed **init** PSNR (dotted) drops with f , while the **trained** PSNR (solid) stays close to the exact-init reference — the gap between the two curves is the amount training recovers.

Initialisation loss

The same recovery shows up in loss space. The perturbed init’s top- k MSE loss on the 500-image train pool — the value the optimiser sees at step 0 — rises with the disturbance rate, from 148.19 at the exact init to 1079.78 at $f = 100\%$ (a $7.3\times$ increase): flat below $\sim 1\%$, then climbing steeply, with a seed spread that widens as more gates are hit. After 1008 steps every perturbed init converges back to essentially the exact-init training loss (120.81 at $f = 100\%$ vs 123.18 at $f = 0$) — training erases the initialisation deficit in the loss just as it does in test PSNR.

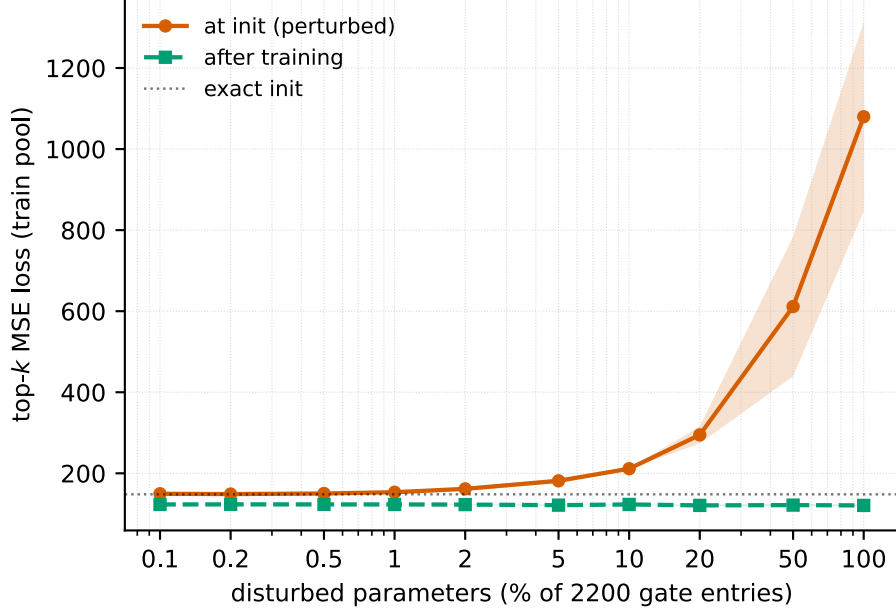


Figure 3: Top- k MSE loss on the 500-image train pool vs. disturbance rate (log-x, linear y): at the perturbed **init** (rising, vermilion) and **after** 1008 training steps (flat, green), mean $\pm \sigma$ over 3 seeds; the dotted line marks the exact-init loss. Training returns every perturbed init to the exact-init loss.

Reading

At $\rho\{=\}.20$ the undisturbed exact init trains to 33.25 dB, and **no amount of disturbance moves that endpoint**: across the entire range — from 0.1% up to **100%** of the 2200 gate entries jittered ($\sigma = 0.1$), i.e. every gate perturbed — trained PSNR stays 33.2– 33.4 dB at $\rho\{=\}.20$ and is equally flat at $\rho\{=\}.01/.05/.10$, never leaving the undisturbed reference by more than ~ 0.2 dB (within seed scatter). The perturbed **init**, by contrast, collapses monotonically — from 30.49 dB at 0.1% to 21.85 dB at 100%, an 8.7 dB fall below the exact init (30.54 dB) that leaves the untrained operator barely above the $\rho\{=\}.20$ floor — yet 1008 steps of top-10% training recover it to 33.44 dB, the exact-init endpoint.

So the exact DCT-IV init sits in a **very wide, flat basin**: a per-gate jitter of $\sigma = 0.1$ applied to **all** its parameters still drains to the same trained optimum. What the trained model depends on is the DCT-IV **topology** plus the top- k objective, not the precise gate values — provided each gate is only locally perturbed. The lever that would break recovery is thus the jitter **magnitude** σ (fixed here), not the disturbed **fraction**: contrast the random-init seed study, where fully Haar-random gates (an $O(1)$, not $\sigma\{=\}0.1$, per-gate displacement) settle into a lower basin below the exact transform.